

Home Bias and Risk Sharing

Sørensen et al. 2007, Empirical Replication and Extension
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Philipp Cremer, Jannis Stagge, Felix Westerhoff

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Outline and Summary

Home bias and international risk sharing: Twin puzzles separated at birth

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Do home bias and risk sharing go hand in hand?

- Risk Sharing: international diversification and smoothing of income streams
- Home Bias: tendency to hold an unproportionally large share of domestic assets

1. Sørensen et al. 2007
 - 1.1 Replication of main cross section and panel regressions
 - 1.2 Extension by time and sample size
2. Eurozone Extension
 - 2.1 Investigation of eurozone home bias
 - 2.2 Channels of home bias reduction on risk sharing in eurozone

- Home bias data
 - External Wealth of Nations Database
 - Stock Market Capitalizations, World Development Indications (World Bank)
- Consumption, GDP and Income data
 - OECD National Accounts

Replication Sørensen et al. 2007

Concept of Home Bias

- same market portfolio of risky assets based on Capital Asset Pricing Model (CAPM)
- countries hold equities/bonds of each country corresponding to each country's share of world market capitalisation
- deviations from this standard are considered a "Home Bias"

Potential Explanations for Home Bias

- currency risk
- transaction costs

$$\text{Equity Home Bias}_i = 1 - \frac{w_i^{\text{foreign}}}{1 - A_i} \quad (1)$$

where:

$$w_i^{\text{foreign}} = \frac{\text{Foreign equity held by country } i}{\text{Country } i\text{'s total equity portfolio}}$$

$$A_i = \frac{\text{Stock market capitalization of country } i}{\text{Stock market capitalization of the world}}$$

and

Total equity portfolio_{*i*} = Market cap_{*i*} + Foreign equity held_{*i*} – Domestic equity held by foreigners_{*i*}

Cross-section Risk Sharing Regressions: Reminder

Setup

Risk sharing regression estimating the **degree of consumption** (Lecture 4, slide 24)

- $$\Delta \log C_{it} - \Delta \log C_t = \alpha + \beta_{C,t}(\Delta \log GDP_{it} - \Delta \log GDP_t) + \varepsilon_{it}$$

Perfect risk sharing

$\beta_{C,t} = 0 \implies$ Consumption perfectly tracks world consumption !

Exercise (Figures 1,2 of reference paper)

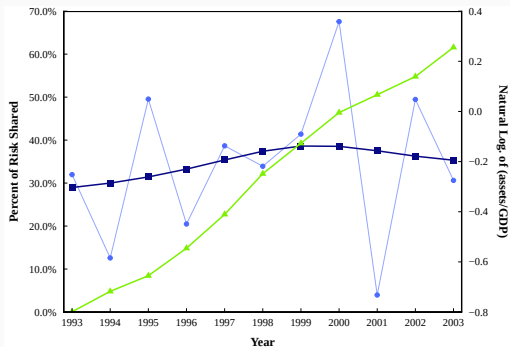
Plot rolling $(1 - \widehat{\beta_{C,t}}) * 100$ against foreign asset holdings over GDP

- Confirmation $\checkmark$: **Increasing trend** of risk sharing and financial openness
- Caveats $\times$: Specific **levels** of risk sharing and overall increase

► Sample

Consumption risk sharing and foreign asset holdings

Replication of Consumption Risk Sharing



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Consumption Risk Sharing Panel Regression

$$\Delta \log C_{it} - \Delta \log C_t = \text{constant} + \kappa (\Delta \log GDP_{it} - \Delta \log GDP_t) + \epsilon_{it} \quad (2)$$

with

$$\kappa = \kappa_0 + \kappa_1(t - \bar{t}) + \kappa_2(EHB_{it} - \overline{EHB_t}) \quad (3)$$

Focus of the analysis on:

$$RiskSharing = 1 - \kappa_0 - \kappa_1(t - \bar{t}) - \kappa_2(EHB_{it} - \overline{EHB_t}) \quad (4)$$

Original Paper

Outcome	Average risk sharing	Trend	Equity Home Bias
Consumption	57 (15.06)	2 (2.09)	−136 (5.48)

Notes: t-values are reported in parantheses.

Replication

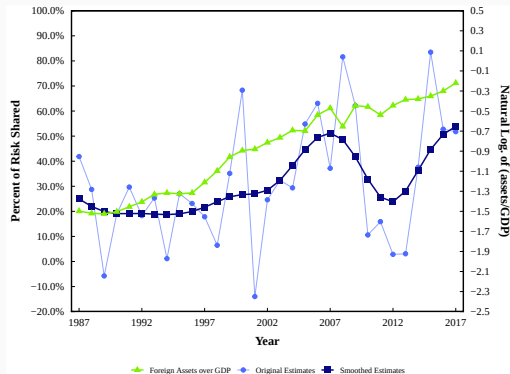
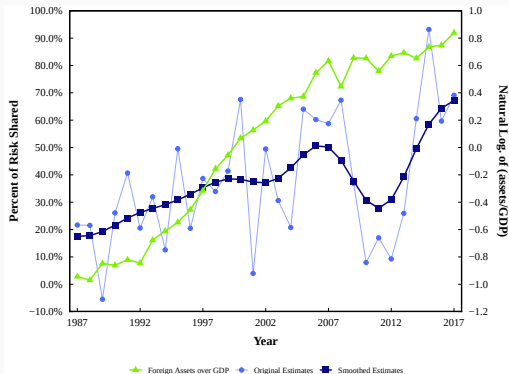
Outcome	Average risk sharing	Trend	Equity Home Bias
Consumption	55 (15.01)	0 (0.52)	−204 (9.07)

Notes: t-values are reported in parantheses.

Within Paper Extension

Sørensen et al. 2007

Extension by Time: Risk Sharing and Equity Home Bias



Extension by time Risk Sharing and Equity Home Bias

Outcome	Average risk sharing	Trend	Equity Home Bias
Consumption	40 (19.61)	0 (0.29)	−81 (7.96)

Notes:t-values are reported in parantheses.

Extension by time and sample size Risk Sharing and Equity Home Bias

Outcome	Average risk sharing	Trend	Equity Home Bias
Consumption	30 (15.28)	0 (0.41)	−84 (8.25)

Notes:t-values are reported in parantheses.

Extensions Eurozone - European Financial Centers

The Puzzle: Euro Area Equity Home Bias

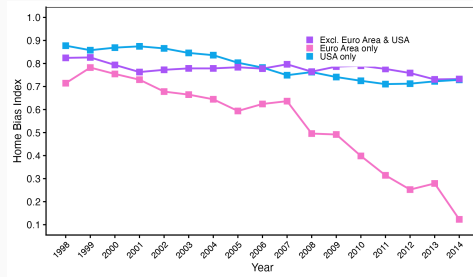
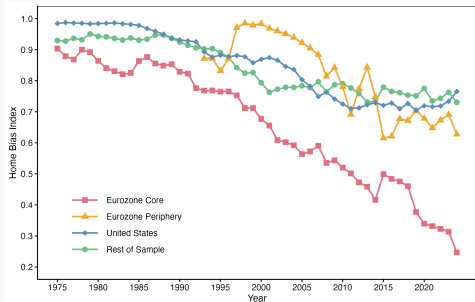
Hypothesis:

- The currency union lowers exchange rate risk and transaction cost
- After 1999, Euro Area countries' EHB should decline faster than non-EA countries
- EA aggregate decline should be less steep than Euro Area countries' EHB decline
- Lower EHB induces stronger within Euro Area Risk sharing

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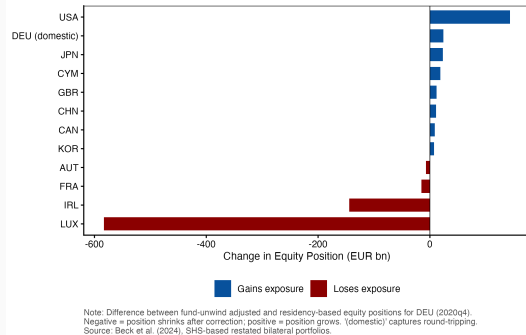
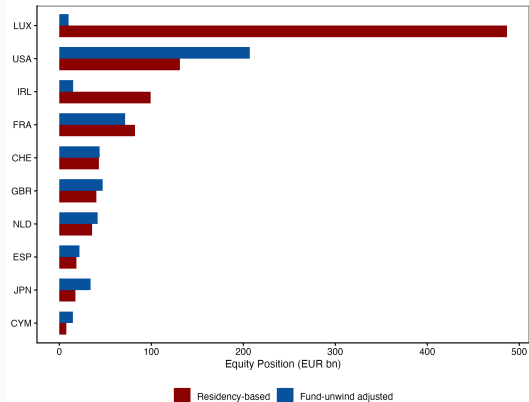
Explanation: Beck et al. (March 2024): **The Geography of Capital Allocation in the Euro Area**

- Luxembourg, Ireland, Netherlands host massive fund industries
- Funds are owned by worldwide investors
- Standard data (e.g. EWN) looks at asset fund owner and origin - not underlying securities

Beck et al. (March 2024)

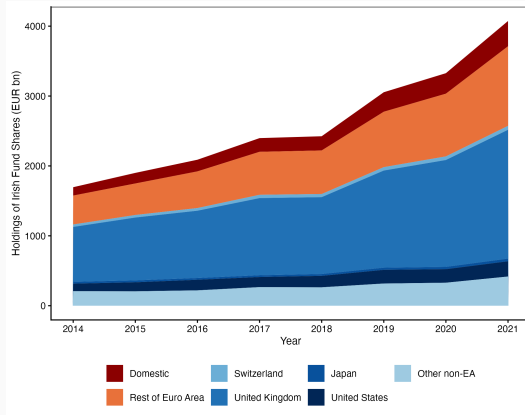
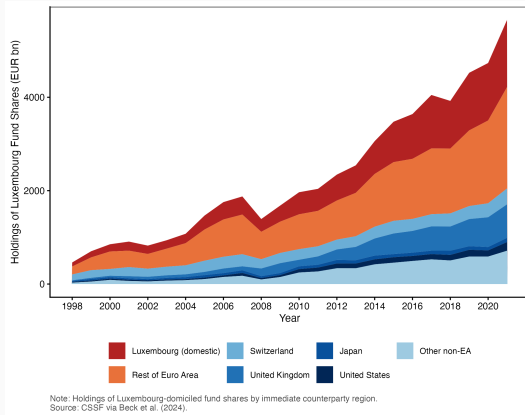
- Use regulatory data and commercial fund holdings data
- “Unwind” fund shares
- Disentangle owners and underlying securities without the funds host country
- Data Available online

Where Does Equity Investment Actually Go? - Example Germany



- Left panel: Germany to foreign country position in standard data vs. corrected data
- Right panel: Net change in position of Germany to (foreign) countries after correction
- Round-Tripping

Who Actually Owns OOFK Funds?



Euro area foreign asset position in standard data strongly inflated

Corrected Home Bias: Equity vs. Bonds (Beck et al., 2024)

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Equity Home Bias

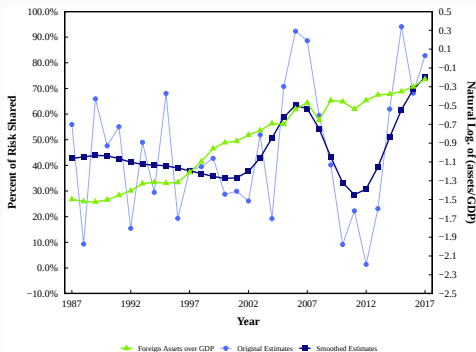
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Bond Home Bias

Extensions Eurozone - Risk Sharing

Consumption Risk Sharing in the Eurozone

- Starting Point: Consumption risk sharing around the introduction of the Euro.
- Finding: Risk sharing seems to have increased around 1999 - but **can this be attributed to a reduction in the Home Bias?**



../output/rs_cs_ext_3b_c.pdf

What if we could isolate the different risk sharing channels? **We can!**

- **Capital Markets Channel**
- Fiscal Transfers Channel
- Credit Market Channel
- Uninsured

This methodology is based on and applied by us on the basis of:

- Asdrubali et al. (**1996**): Seminal paper (> 1000 citations)
- Hosseini and Gervais (**March 2026**): Replicable methodology (used for Eurozon sample)

Consumption Risk Sharing Channels in the Eurozone

And again, relying on rolling cross-section estimates

- Finding: After the GFC, risk sharing via credit markets markedly increased

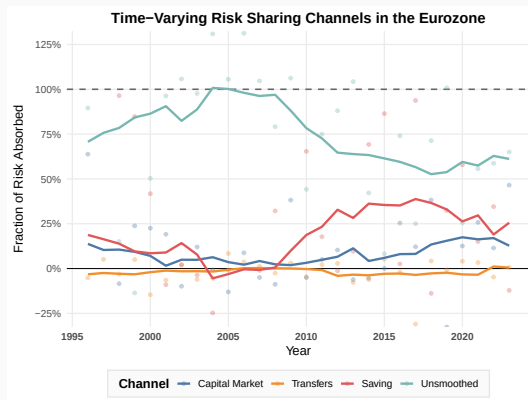


Figure 3: Year-by-year cross-sectional estimates (10-year centred moving averages)

- Home Bias and Risk Sharing are **negatively** associated (causal relationship?)
- Finding in Sørensen et al. (2007) **robust** against various sample extensions (Note: **specification sensitivity!**)
- Erronoues Data quality leads to EHB **downward distortion**
- Debt Home Bias declined stronger in Euro Area compared to US
- Capital Markets **gradually more important** for risk sharing (after GFC)
- Consumption Smoothing remains at **low levels** compared to US
- Potential Cause: **Incomplete European Integration**, no Capital Markets/Banking Union or Fiscal Transfers

-  Asdrubali, P., B. E. Sorensen, and O. Yosha (1996). **“Channels of Interstate Risk Sharing: United States 1963-1990”**. In: *The Quarterly Journal of Economics* 111.4, pp. 1081–1110.
-  Beck, R. et al. (March 2024). **The Geography of Capital Allocation in the Euro Area**. Working Paper 32275. National Bureau of Economic Research. DOI: [10.3386/w32275](https://doi.org/10.3386/w32275). URL: <http://www.nber.org/papers/w32275>.
-  Hosseini, R. and M. Gervais (March 2026). **Risk Sharing across U.S. States and EU Countries**. Working Paper. SSRN. URL: https://roozbeh-hosseini.github.io/papers/risk_sharing_states.pdf.
-  Sørensen, B. E. et al. (2007). **“Home bias and international risk sharing: Twin puzzles separated at birth”**. In: *Journal of International Money and Finance* 26.4. Financial Globalization and Integration, pp. 587–605.

Appendix: Equity Home Bias Comparison

Paper (Sørensen et al. 2007)

	Foreign eq. in portf. (%)		Equity Home Bias	
	1993	2003	1993	2003
Australia	11.26	17.20	0.89	0.82
Austria	13.03	61.14	0.87	0.39
Belgium	45.94	50.16	0.54	0.50
Canada	26.00	30.27	0.73	0.69
Denmark	17.14	36.58	0.83	0.63
Finland	1.66	35.02	0.98	0.65
France	12.86	28.15	0.87	0.71
Germany	23.75	44.70	0.75	0.54
Greece	4.27	4.30	0.96	0.96
Iceland	—	—	—	—
Ireland	—	—	—	—
Italy	21.25	41.84	0.79	0.57
Japan	3.59	9.97	0.95	0.89
Mexico	1.16	11.45	0.99	0.89
Netherlands	40.00	62.01	0.59	0.37
New Zealand	7.10	35.10	0.93	0.65
Norway	16.70	51.45	0.83	0.48
Portugal	14.20	31.98	0.86	0.68
Spain	6.31	13.97	0.94	0.86
Sweden	14.94	41.60	0.85	0.58
Switzerland	40.13	47.52	0.59	0.51
Turkey	1.74	2.37	0.98	0.98
UK	23.16	29.51	0.75	0.68
US	10.25	14.32	0.84	0.74
Average	16.20	31.85	0.83	0.67

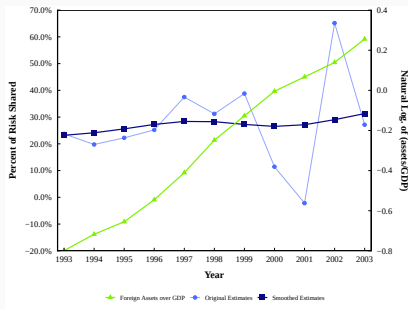
Our Replication

Table 1: Equity Home Bias 1993 and 2003

Country	1993		2003	
	Foreign equity in portfolio (%)	Equity Home Bias	Foreign equity in portfolio (%)	Equity Home Bias
Australia	11.33	0.89	17.23	0.82
Austria	14.12	0.86	59.50	0.40
Belgium	45.97	0.54	50.17	0.50
Canada	11.16	0.88	27.21	0.72
Denmark	17.85	0.82	36.57	0.63
Finland	1.66	0.98	NA	NA
France	12.86	0.87	28.23	0.70
Germany	24.87	0.74	44.57	0.54
Greece	NA	NA	4.62	0.95
Iceland	NA	NA	NA	NA
Ireland	NA	NA	NA	NA
Italy	22.07	0.78	45.70	0.53
Japan	3.82	0.95	10.29	0.89
Mexico	1.16	0.99	12.48	0.87
Netherlands	40.03	0.59	68.53	0.30
New Zealand	2.80	0.97	42.60	0.57
Norway	22.23	0.78	50.99	0.49
Portugal	NA	NA	37.38	0.63
Spain	5.85	0.94	13.05	0.87
Sweden	16.05	0.84	41.60	0.58
Switzerland	40.28	0.59	47.51	0.51
Turkey	1.83	0.98	2.33	0.98
United Kingdom	23.18	0.75	36.17	0.61
United States	10.03	0.84	14.33	0.73
Average	16.46	0.83	32.91	0.66

Appendix: Volatility of cross-section estimates

The cross-section risk sharing regressions are highly volatile to the **price measurement**.



2000 as base year

../output/rs_cs_rebased_c.pdf

Chained-linked volumes, rebased to 2020¹

Figure 4: Consumption Risk Sharing (similar for GNI)

¹ "Chained [linked volumes] better reflect changes in aggregates than constant prices" (Source: OECD)

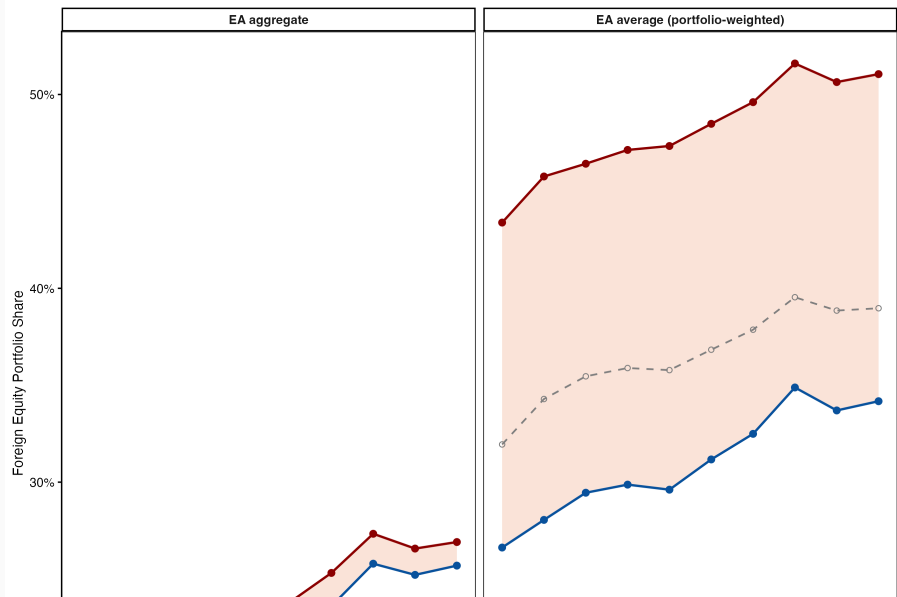
Appendix: Volatility of cross-section estimates

The sample includes Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Greece, Iceland, Ireland, Italy, Japan, Mexico, the Netherlands, New Zealand, Norway, Portugal, Spain, Sweden, Switzerland, Turkey, the UK, and the United States.

The sample is complete for foreign asset holdings over GDP and consumption and does not include Turkey for GNI (data availability).

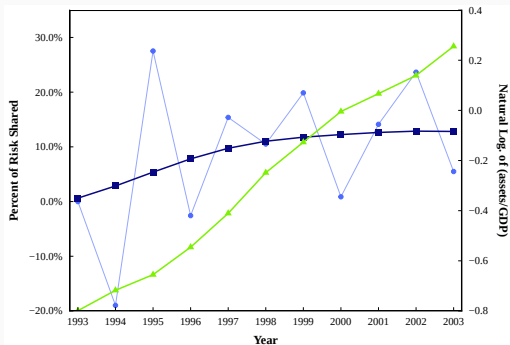
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Appendix: Corrected Foreign Equity Portfolio Shares



Appendix: Consumption risk sharing and foreign asset holdings

Replication of Income Risk Sharing



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Appendix: Risk Sharing and Foreign Asset Holdings relative to GDP, OECD Sample 1993–2003

Original Paper

Outcome	Measure	Equity	Debt	FDI	Equity + Debt	All Assets
Income	Average risk sharing	6 (2.74)	5 (2.53)	3 (1.70)	6 (2.85)	6 (2.75)
	Trend	0 (0.77)	0 (0.34)	0 (0.47)	0 (0.37)	0 (0.47)
	Asset holdings	5 (4.50)	9 (4.46)	3 (2.70)	9 (4.65)	8 (4.35)
Consumption	Average risk sharing	51 (13.34)	45 (11.56)	51 (13.39)	47 (11.95)	48 (12.34)
	Trend	1 (0.83)	0 (0.25)	0 (0.09)	0 (0.33)	0 (0.45)
	Asset holdings	11 (4.02)	7 (1.64)	9 (3.79)	9 (2.26)	11 (2.74)

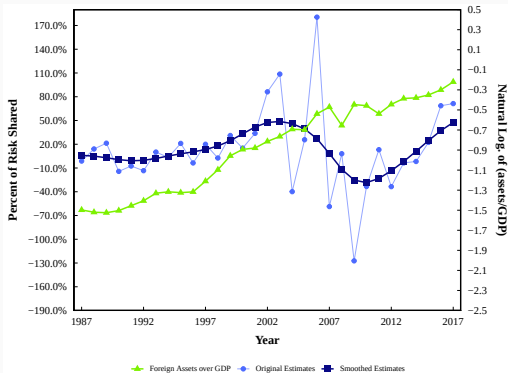
Notes: t-values are reported in parantheses.

Replication

Outcome	Measure	Equity	Debt	FDI	Equity + Debt	All Assets
Income	Average risk sharing	15.064 (4.145)	15.784 (4.211)	9.675 (2.788)	15.997 (4.253)	15.709 (4.262)
	Trend	1.202 (1.209)	0.715 (0.719)	1.685 (1.634)	0.733 (0.729)	1.069 (1.062)
	Asset holdings	12.180 (5.190)	17.747 (5.154)	11.726 (3.333)	17.583 (5.046)	20.515 (5.142)
Consumption	Average risk sharing	56.614 (14.769)	49.948 (12.157)	51.790 (12.591)	52.469 (12.928)	54.959 (13.589)
	Trend	0.942 (0.903)	0.172 (0.153)	0.455 (0.407)	0.355 (0.322)	0.497 (0.457)
	Asset holdings	17.834 (8.796)	21.312 (6.009)	14.252 (6.160)	22.576 (6.814)	24.189 (7.348)

Notes:t-values are reported in parantheses.

Appendix: Income Risk Sharing in the Eurozone



Eurozone only

../output/rs_cs_ext_3b_i.pdf

Non-Eurozone

Figure 6: Income Risk Sharing cross-sectional estimates

Appendix: Consumption Risk Sharing Channels in the Eurozone

Applying this methodology, on the basis of Hosseini and Gervais (March 2026):

- Finding: 80% of output volatility unsmoothed (US interstate risk sharing), 6% via capital markets, 11% via credit markets

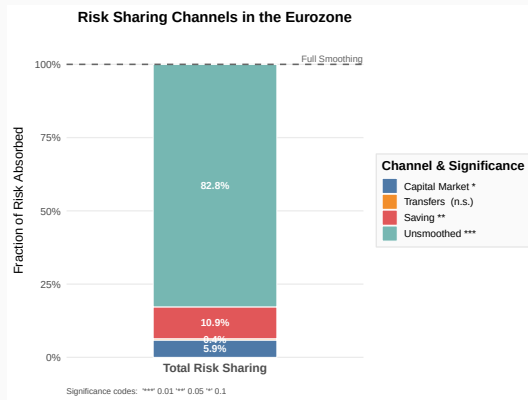
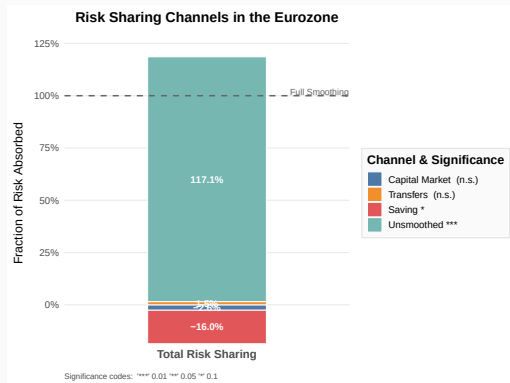


Figure 7: Risk Sharing Channels in the Eurozone: Panel Estimation 1995 – 2023

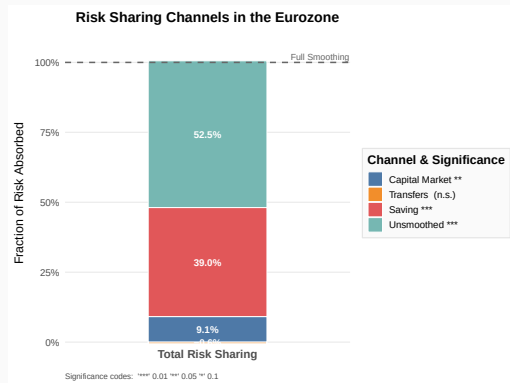
Appendix: Consumption Risk Sharing Channels

After the time sample between before and after the GFC:

- Finding: risk sharing improves over time through capital and credit markets.



1995 – 2007



2010 – 2023

Figure 8: Risk Sharing Channels across different time periods